Truthmaker Semantics as Metaphysics

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Introduction - truthmaker semantics

State spaces

We'll be considering some notions of "truthmaker model" for a simple formal language. Each model is built on a *state space* $\langle S, \sqsubseteq \rangle$: a *complete semilattice*, i.e. a partial order in which every nonempty $p \subseteq S$ has a least upper bound.

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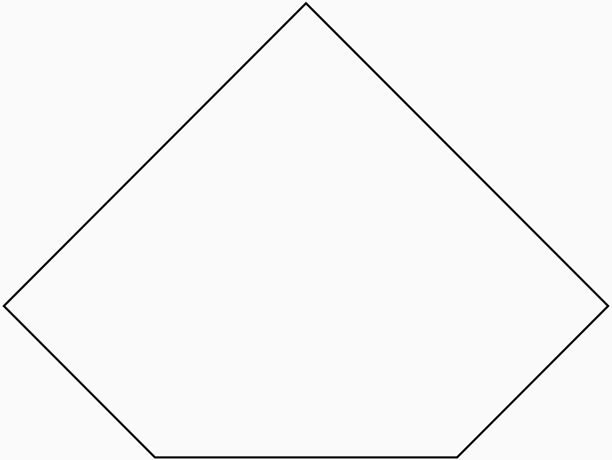
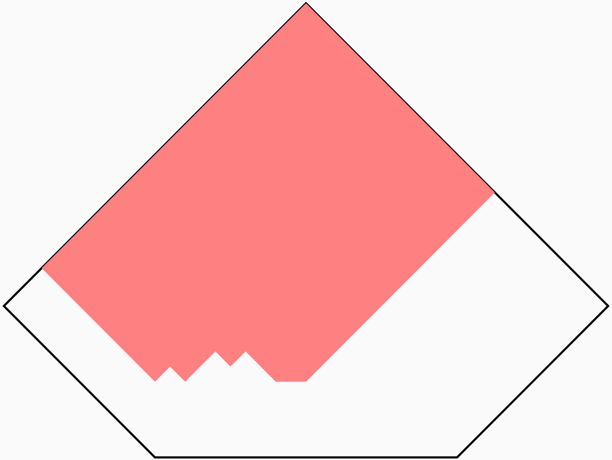
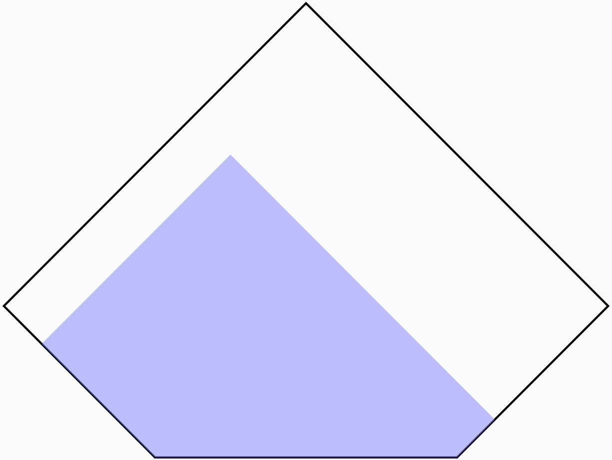
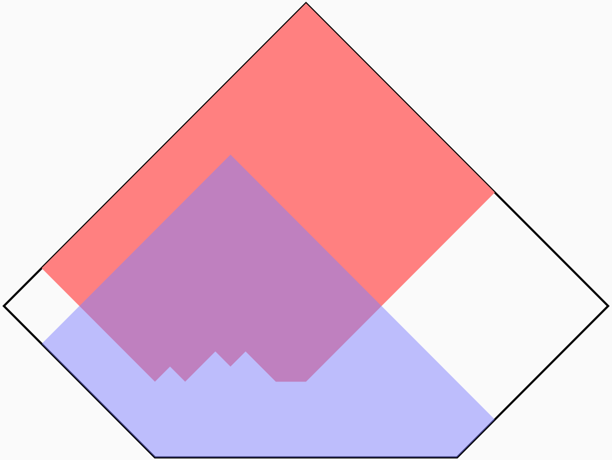
Definitions: ($\langle S, \sqsubseteq \rangle$ is a complete semilattice, $s, s' \in S$, $p, q \subseteq S$):

- $s \sqcup s'$ (the **fusion** of s and s') $\coloneqq \sup\{s, s'\}$
- $p \mathbin{\dot\sqcup} q$ (the **pointwise fusion** of p and q) $\coloneqq \{s \sqcup t \mid s \in p, t \in q\}$
- $p^{\uparrow}$ (the **upward closure** of p) $\coloneqq \{s : s' \sqsubseteq s \text{ for some } s' \in p\}$
- $p^{\downarrow}$ (the **downward closure** of p) $\coloneqq \{s : s \sqsubseteq s' \text{ for some } s' \in p\}$
- p^*_* (the **regular closure** of p) $\coloneqq p^{\uparrow} \cap \{\sup p\}^{\downarrow}$
- p is **regular** $\coloneqq p = p^*_*$.









Varieties of truth-maker semantics: three choice points

In defining "M is a truthmaker model based on $\langle S, \sqsubseteq \rangle$", we face three choice points:

(A) In *unilateral* truthmaker models, the interpretation $[\![\phi]\!]$ of a sentence ϕ is a *subset* of S (the set of "verifiers" for ϕ).

(B) In *bilateral* truthmaker models, $[\![\phi]\!]$ is an *ordered pair* of subsets of S—the "verifiers" and the "falsifiers" of ϕ.

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(1) In *regular* models, the verifier/falsifier sets must be *regular*.

(2) In *irregular* models they need not be.

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(1) In *regular* models, the verifier/falsifier sets must be *regular*.

(2) In *irregular* models they need not be.

(i) In *witnessed* models the verifier/falsifier sets cannot be empty.

(ii) In *vacuity-tolerant* models they may be.

This yields eight versions. (Further subdivisions are possible!)

Truthmaker models for a propositional language

The language $\mathcal{L}_{\textbf{prop}}$

$$\phi \coloneqq v \mid \neg\phi \mid \phi \wedge \psi \mid \phi \vee \psi \qquad (v \in \textbf{Var})$$

Abbreviations:

$$\phi \to \psi \coloneqq \neg\phi \vee \psi$$
$$\phi \leftrightarrow \psi \coloneqq (\phi \to \psi) \wedge (\psi \to \phi)$$

Let's consider how the eight variants assign denotations to formulae of $\mathcal{L}_{\textbf{prop}}$.

Irregular semantics for $\mathcal{L}_{\textbf{prop}}$

The basic idea of the **irregular unilateral semantics** is to have:

$$[\![\phi \vee \psi]\!] = [\![\phi]\!] \cup [\![\psi]\!]$$
$$[\![\phi \wedge \psi]\!] = [\![\phi]\!] \mathbin{\dot\sqcup} [\![\psi]\!]$$

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In the **irregular bilateral semantics**, by contrast, we can put

$$[\![\neg\phi]\!] = \langle \pi_2[\![\phi]\!], \pi_1[\![\phi]\!] \rangle$$

while changing the clauses for $\vee$ and $\wedge$ to:

$$[\![\phi \wedge \psi]\!] = \langle \pi_1[\![\phi]\!] \uplus \pi_1[\![\psi]\!], \pi_2[\![\phi]\!] \cup [\![\psi]\!] \rangle$$
$$[\![\phi \vee \psi]\!] = \langle \pi_2[\![\phi]\!] \cup \pi_2[\![\psi]\!], \pi_1[\![\phi]\!] \uplus \pi_1[\![\psi]\!] \rangle$$

Regular conjunction and disjunction

Since the union and pointwise fusion of two regular sets can be irregular, we can't use these clauses in the regular semantics. Instead, we can take regular closures.

In the **regular unilateral** semantics, we set

$$[\![\phi \wedge \psi]\!] = [\![\phi]\!] \mathbin{\dot\wedge} [\![\psi]\!] \text{ where } p \mathbin{\dot\wedge} q \coloneqq (p \sqcup q)^*_* = (p^\uparrow \cap q^\uparrow) \cap (\sup(p \cup q))^\downarrow$$

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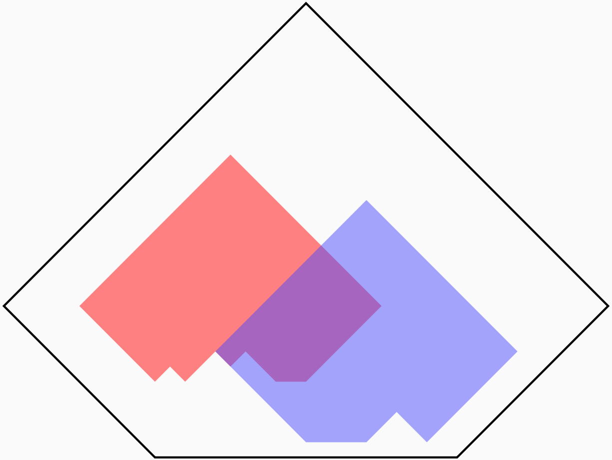
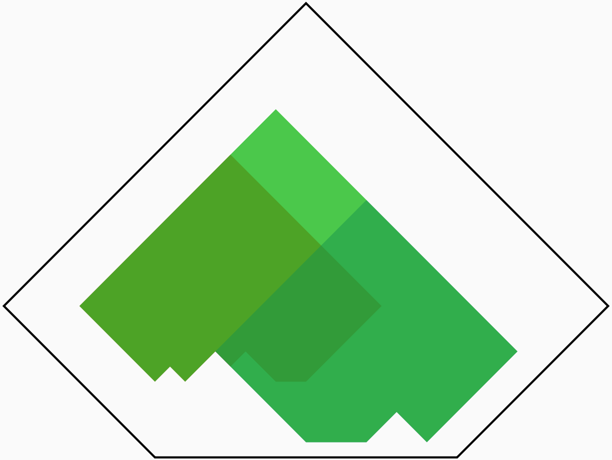
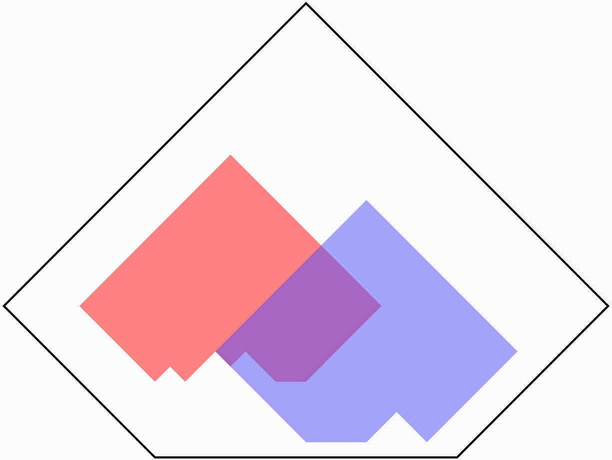
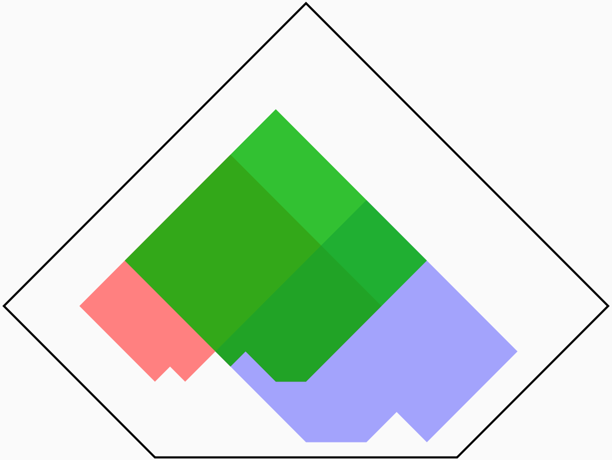
As in the irregular case, we'll need to add some extra structure to treat negation.









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$$[\![\phi \vee \psi]\!] = \langle \pi_2[\![\phi]\!] \mathbin{\dot{\vee}} \pi_2[\![\psi]\!], \pi_1[\![\phi]\!] \mathbin{\dot{\wedge}} \pi_1[\![\psi]\!] \rangle$$

Some characteristic equivalences and non-equivalences

All eight versions guarantee the *Commutativity* and *Associativity* of $\wedge$ and $\vee$:

$$[\![\phi \wedge \psi]\!] = [\![\psi \wedge \phi]\!] \qquad [\![\phi \vee \psi]\!] = [\![\psi \vee \phi]\!]$$

$$[\![\phi \wedge (\psi \wedge \chi)]\!] = [\![(\phi \wedge \psi) \wedge \chi]\!] \qquad [\![\phi \vee (\psi \vee \chi)]\!] = [\![(\phi \vee \psi) \vee \chi]\!]$$

All eight are compatible with there being failures of *Absorption*:

$$[\![\phi \wedge (\phi \vee \psi)]\!] \stackrel{?}{=} [\![\phi]\!] \qquad [\![\phi \vee (\phi \wedge \psi)]\!] \stackrel{?}{=} [\![\phi]\!]$$

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Meanwhile, the status of *Idempotence* and *Distributivity* varies between different varieties of model:

$$[\![\phi \vee \phi]\!] \stackrel{?}{=} [\![\phi]\!] \qquad [\![\phi \wedge \phi]\!] \stackrel{?}{=} [\![\phi]\!]$$

$$[\![\phi \wedge (\psi \vee \chi)]\!] \stackrel{?}{=} [\![(\phi \wedge \psi) \vee (\phi \wedge \chi)]\!] \qquad [\![\phi \vee (\psi \wedge \chi)]\!] \stackrel{?}{=} [\![(\phi \vee \psi) \wedge (\phi \vee \chi)]\!]$$

Idempotence and Distributivity

	Unilateral				Bilateral			
	Irregular		Regular		Irregular		Regular	
	V	W	V	W	V	W	V	W
$[\![\phi \wedge \phi]\!] = [\![\phi]\!]$								
$[\![\phi \vee \phi]\!] = [\![\phi]\!]$								
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$[\![\phi \wedge \phi]\!] = [\![\phi]\!]$								
$[\![\phi \vee \phi]\!] = [\![\phi]\!]$	✓	✓	✓	✓				
$[\![\phi \wedge (\psi \vee \chi)]\!] = [\![(\phi \wedge \psi) \vee (\phi \wedge \chi)]\!]$	✓	✓	✓	✓				
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$[\![\phi \wedge \phi]\!] = [\![\phi]\!]$	×	×	✓	✓				
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$[\![\phi \wedge \phi]\!] = [\![\phi]\!]$	×	×	✓	✓				
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$[\![\phi \wedge (\psi \vee \chi)]\!] = [\![(\phi \wedge \psi) \vee (\phi \wedge \chi)]\!]$	✓	✓	✓	✓				
$[\![\phi \vee (\psi \wedge \chi)]\!] = [\![(\phi \vee \psi) \wedge (\phi \vee \chi)]\!]$	×	×						

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	Irregular		Regular		Irregular		Regular	
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$[\![\phi \vee \phi]\!] = [\![\phi]\!]$	✓	✓	✓	✓				
$[\![\phi \wedge (\psi \vee \chi)]\!] = [\![(\phi \wedge \psi) \vee (\phi \wedge \chi)]\!]$	✓	✓	✓	✓				
$[\![\phi \vee (\psi \wedge \chi)]\!] = [\![(\phi \vee \psi) \wedge (\phi \vee \chi)]\!]$	×	×	×	✓				

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	V	W	V	W	V	W	V	W
$[\![\phi \wedge \phi]\!] = [\![\phi]\!]$	×	×	✓	✓	×	×	✓	✓
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$[\![\phi \vee (\psi \wedge \chi)]\!] = [\![(\phi \vee \psi) \wedge (\phi \vee \chi)]\!]$	×	×	×	✓	×	×	×	✓
$[\![\neg(\phi \vee \psi)]\!] = [\![\neg\phi \wedge \neg\psi]\!]$					✓	✓	✓	✓
$[\![\neg(\phi \wedge \psi)]\!] = [\![\neg\phi \vee \neg\psi]\!]$					✓	✓	✓	✓
$[\![\neg\neg\phi]\!] = [\![\phi]\!]$					✓	✓	✓	✓

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$[\![\phi \vee \phi]\!] = [\![\phi]\!]$	✓	✓	✓	✓	×	×	✓	✓
$[\![\phi \wedge (\psi \vee \chi)]\!] = [\![(\phi \wedge \psi) \vee (\phi \wedge \chi)]\!]$	✓	✓	✓	✓	×	×	×	✓
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$[\![\neg(\phi \wedge \psi)]\!] = [\![\neg\phi \vee \neg\psi]\!]$	(×)	(×)	(✓)	(✓)	✓	✓	✓	✓
$[\![\neg\neg\phi]\!] = [\![\phi]\!]$	(×)	(×)	(×)	(×)	✓	✓	✓	✓

Truthmaker Semantics as Metaphysics

As metaphysicians, we think we can *understand* sentences like $\forall p(p = p \wedge p)$ and ask whether they are true or not.

- Not "true in models of such-and-such sort", or "true on such-and-such conception of the proposition". Just *true*.
- There are various candidate ways of making the relevant claim in English: we might try 'For things to be a given way is for them to both be that way and be that way', or 'Every proposition is the proposition that it is both true and true', or maybe 'Every state of affairs is the conjunction of that state with itself'. We'll help ourselves to these ways of talking without taking a stand on whether they are really accurate given the intricacies of English.

How might the mathematical definitions that constitute truthmaker semantics be of interest for metaphysics?

First, one might think that some class of truthmaker models gives an accurate account of reality in the sense that every sentence true in every model in the class is in fact true.

- This will of course require introducing a notion of "truth in a model". Moreover, if we are interested in sentences like $\forall p(p \wedge p = p)$, we'll need it to apply not just to $\mathcal{L}_{\mathbf{prop}}$ but to a language with propositional quantification and identity.

Second, one might (more ambitiously) regard some class of models not merely as an accurate guide to questions of propositional identity, but as a good picture of the structure of reality on their own.

If so, one might use the terminology of TMS (‘state’, ‘verify’, ‘falsify’,...) in an unrelativized way, to talk about the supposed structure in reality corresponding to the structure of such models.

- For example, one will take oneself to understand not just model-relative questions like ‘How many states in such-and-such model are verifiers for the sentence “Snow is white”?’, but unrelativized questions like ‘How many states are verifiers for snow being white?’

Opponents might question the intelligibility of the novel terminology (‘state’, etc.),

One way to defuse such worries is to provide *definitions* of the novel terminology in more familiar terms.

We will show how at least in some versions of truthmaker semantics, one can define ‘state’, ‘verify’, etc. in terms of propositional quantification and identity.

The language $\mathcal{L}$

$$\phi \coloneqq p \mid \neg\phi \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi = \psi \mid \forall p\, \phi \mid \exists p\, \phi \qquad (p \in \mathbf{Var})$$

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Abbreviations:

$$\phi \leq_{\vee} \psi \coloneqq \psi = \psi \vee \phi \qquad \phi \leq_{\vee\exists} \psi \coloneqq \exists p(\psi = p \vee \phi)$$

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In the below, γ and δ stand for formulae in which some variable $\hat{v}$ is decorated with a hat, $\gamma[\phi]$ and $\delta[\phi]$ are the results of substituting ϕ for $\hat{v}$.

$$\mathbf{UB}_{\leq_{\#}}(\phi, \gamma) \coloneqq \forall p(\gamma[p] \rightarrow p \leq_{\#} \phi)$$

$$\mathbf{Sup}_{\leq_{\#}}(\phi, \gamma) \coloneqq \mathbf{UB}_{\leq_{\#}}(\phi, \gamma) \wedge \forall q(\mathbf{UB}_{\leq_{\#}}(q, \gamma) \rightarrow \phi \leq_{\#} q)$$

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$$\mathbf{Atom}_{\leq_{\#}}(\phi) \coloneqq \forall p(p \leq_{\#} \phi \to (\neg(\phi \leq_{\#} p) \leftrightarrow \forall q(p \leq_{\#} q))$$

The language $\mathcal{L}^-$

This is like $\mathcal{L}$, but with the restriction that $\phi = \psi$ is only well-formed when ϕ and ψ are formulae of $\mathcal{L}_{\mathbf{prop}}$ (i.e., no identity or quantifiers under $=$).

Since we we are interested in extracting metaphysical mileage from the core features of truthmaker semantics, namely its treatment of the Boolean connectives, $\mathcal{L}^-$ will suffice for our purposes today (although we think $\mathcal{L}$ is perfectly intelligible).

A precedent: possible worlds semantics

Possible worlds models

A PW-model is a tuple $\langle W, @, [\![\cdot]\!] \rangle$, with $@ \in W$ and $[\![\cdot]\!]^{\cdot} : \mathcal{P}(W)^{\mathbf{Var}} \to \mathcal{P}(W)^{\mathcal{L}}$, such that:

Variables $[\![p]\!]_g = g(p)$

Conjunction $[\![\phi \wedge \psi]\!]_g = [\![\phi]\!]_g \cap [\![\psi]\!]_g$

Disjunction $[\![\phi \vee \psi]\!]_g = [\![\phi]\!]_g \cup [\![\psi]\!]_g$

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Minimal $\forall$ $@ \in [\![\forall p \phi]\!]$ iff $@ \in [\![\phi]\!]_{g[p \mapsto X]}$ for all $X \subseteq W$

Minimal $\exists$ $@ \in [\![\exists p \phi]\!]$ iff $@ \in [\![\phi]\!]_{g[p \mapsto X]}$ for some $X \subseteq W$

Minimal $=$ $@ \in [\![\phi = \psi]\!]$ iff $[\![\phi]\!]_g = [\![\psi]\!]_g$

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ϕ is **true in** a PW-model iff $@ \in [\![\phi]\!]_g$ for all g.

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(There are well-known ways of strengthening Minimal $\forall$, $\exists$, and $=$, but they are irrelevant to the truth of sentences of $\mathcal{L}^-$.)

The theory Internal PW

Extend $\mathcal{L}^-$ to a two-sorted language $\mathcal{L}_{\mathbf{PW}}$ by adding *world variables* $v, w, \ldots$, not allowed to occur under identity.

When ϕ is a formula and w is a world variable, $w \rhd \phi$ ("ϕ is true at w") and $\mathbf{Obt}\, w$ are formulae, as are $\forall w \phi$ and $\exists w \phi$.

Internal PW adds the following axioms to classical logic in this signature:

Uniqueness	$\forall w(w \rhd p \leftrightarrow w \rhd q) \rightarrow p = q$
Existence	$\exists p \forall w(w \rhd p \leftrightarrow \gamma[w])$
Truth	$\exists w(\mathbf{Obt}\, w \wedge w \rhd p) \leftrightarrow p$
I-Conjunction	$w \rhd (p \wedge q) \leftrightarrow (w \rhd p \wedge w \rhd q)$
I-Disjunction	$w \rhd (p \vee q) \leftrightarrow (w \rhd p \vee w \rhd q)$
I-Negation	$w \rhd \neg p \leftrightarrow \neg (w \rhd p)$

The reduction

Strategy: use singleton subsets of W as surrogates for elements of W.

This works because we can pick out singletons in the object language: $\mathbf{Atom}_{\leq_\vee}(\phi)$ is true on g iff $[\![\phi]\!]_g = \{w\}$ for some w.

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We define a mapping $|\cdot|$ from $\mathcal{L}_{\mathbf{PW}}$ to $\mathcal{L}$:

$$|w \rhd \phi| \coloneqq w \leq_\vee |\phi| \qquad |\mathbf{Obt}\, w| \coloneqq w$$

$$|\forall w \phi| \coloneqq \forall w(\mathbf{Atom}_{\leq_\vee} w \to |\phi|) \qquad |\exists w \phi| \coloneqq \exists w(\mathbf{Atom}_{\leq_\vee} w \wedge |\phi|)$$

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$$|\phi \wedge \psi| \coloneqq |\phi| \wedge |\psi| \qquad |\phi \vee \psi| \coloneqq |\phi| \vee |\psi|$$

$$|\neg\phi| \coloneqq \neg|\phi| \qquad |\phi = \psi| \coloneqq |\phi| = |\psi|$$

$$|\forall p \phi| \coloneqq \forall p|\phi| \qquad |\exists p \phi| \coloneqq \exists p|\phi|$$

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$$
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|\phi \wedge \psi| &\coloneqq |\phi| \wedge |\psi| & |\phi \vee \psi| &\coloneqq |\phi| \vee |\psi| \\
|\neg \phi| &\coloneqq \neg |\phi| & |\phi = \psi| &\coloneqq |\phi| = |\psi| \\
|\forall p \phi| &\coloneqq \forall p |\phi| & |\exists p \phi| &\coloneqq \exists p |\phi|
\end{aligned}
$$

Fact

$|\phi|$ is true in every PW-model whenever ϕ is a theorem of Internal PW.

A more perspicuous axiomatization of |Internal PW|

It is worth looking for some simpler, more natural axioms equivalent to |Internal PW|. We can start with **B**, the result of adding the following (highly redundant) set of **Boolean identities** to classical logic:

Com	$p \wedge q = q \wedge p$	$p \vee q = q \vee p$
Assoc	$p \wedge (q \wedge r) = (p \wedge q) \wedge r$	$p \vee (q \vee r) = (p \vee q) \vee r$
Id	$p \wedge p = p$	$p \vee p = p$
Dist	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
Abs	$p \wedge (p \vee q) = p$	$p \vee (p \wedge q) = p$
Comp	$p \wedge \neg p = q \wedge \neg q$	$p \vee \neg p = q \vee \neg q$
DeM	$\neg(p \wedge q) = \neg p \vee \neg q$	$\neg(p \vee q) = \neg p \wedge \neg q$
Invol	$\neg\neg p = p$	

A more perspicuous axiomatization of $|$Internal PW$|$

On top of the Boolean identities, we can add one member from each of the following groups of axioms:

$\vee$-**Comp**	$\exists p\, \mathbf{Sup}_{\leq_\vee}(p, \gamma)$
$\wedge$-**Comp**	$\exists p\, \mathbf{Sup}_{\leq_\wedge}(p, \gamma)$
Atomicity	$\forall q(p \leq_\vee q) \vee \exists q(\mathbf{Atom}_{\leq_\vee} q \wedge q \leq_\vee p)$
Atomisticity	$\mathbf{Sup}_{\leq_\vee}(p, \mathbf{Atom}_{\leq_\vee} \hat{q} \wedge \hat{q} \leq_\vee p)$
Actuality	$\exists p(p \wedge \forall q(q \to q \leq_\wedge p))$
Anti-Actuality	$\exists p(\neg p \wedge \forall q(\neg q \to q \leq_\vee p))$
True Atom	$\exists p(p \wedge \mathbf{Atom}_{\leq_\vee} p)$

Call the resulting theory (which includes all seven axioms) “B+”.

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Call the resulting theory (which includes all seven axioms) "B+".

Fact

B+ is equivalent to $|$Internal PW$|$.

Regular witnessed lhunilateral truthmaker models

Regular, unilateral truthmaker models

$\mathcal{R}(S, \sqsubseteq) \coloneqq \{p \subseteq S : p \text{ is regular and non-empty}\}$

A **RU-model** is a tuple $\mathfrak{M} = \langle S, \sqsubseteq, O, \sim, [\![\cdot]\!]\rangle$, where $\langle S, \sqsubseteq\rangle$ is a complete semilattice,

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Obtaining Parts $O = O^{\downarrow}$
Obtaining Closure $\sup O \in O$

$\Vdash p$ (p is **true in** $\mathfrak{M}$) iff $O \cap p \neq \varnothing$. ϕ is **true in** $\mathfrak{M}$ iff $\Vdash [\![\phi]\!]_g$ for all g.

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Obtaining Parts	$O = O^{\downarrow}$
Obtaining Closure	$\sup O \in O$

Non-Contradiction	$\nVdash \sim t$ when $t \in O$
Bivalence	$\Vdash \sim t$ when $t \notin O$
Sup Exclusion	$\sim \sup p = \dot{\bigvee}\{\sim s : s \in p\}$

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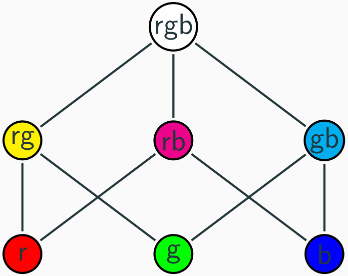
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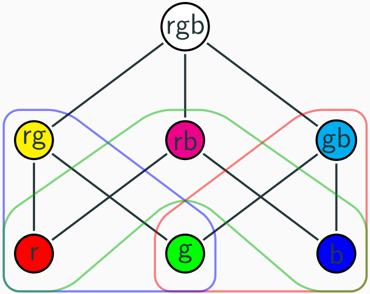
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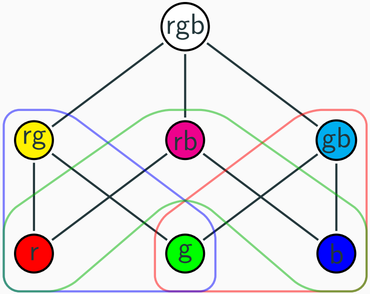
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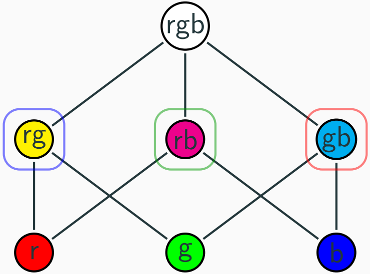
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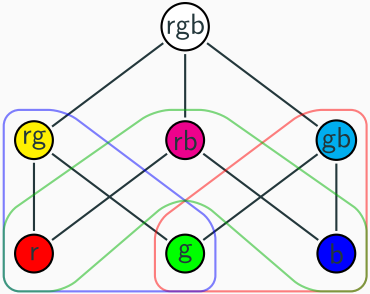

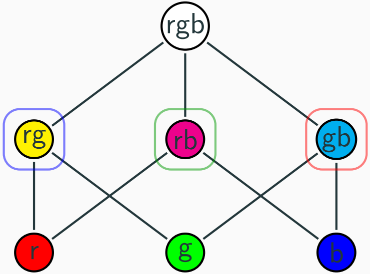

s is *consistent* iff whenever $t \sqsubseteq s$ and $t' \in {\sim}t$, $t' \not\sqsubseteq s$.

s is *complete* iff for every t, either $t \sqsubseteq s$ or $t' \sqsubseteq s$ for some $t' \in {\sim}t$.

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Obtaining Parts	$O = O^{\downarrow}$
Obtaining Closure	$\sup O \in O$

Non-Contradiction	$\nVdash {\sim}t$ when $t \in O$
Bivalence	$\Vdash {\sim}t$ when $t \notin O$
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		Variables	$[\![p]\!]_g = g(p)$
Obtaining Parts	$O = O^{\downarrow}$	**Conjunction**	$[\![\phi \wedge \psi]\!]_g = [\![\phi]\!]_g \mathbin{\dot\wedge} [\![\psi]\!]_g$
Obtaining Closure	$\sup O \in O$	**Disjunction**	$[\![\phi \vee \psi]\!]_g = [\![\phi]\!]_g \mathbin{\dot\vee} [\![\psi]\!]_g$
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Obtaining Parts	$O = O^{\downarrow}$	**Conjunction**	$[\![\phi \wedge \psi]\!]_g = [\![\phi]\!]_g \curlywedge [\![\psi]\!]_g$
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		Negation	$[\![\neg\phi]\!] = \bigwedge\{\sim s : s \in [\![\phi]\!]_g\}$
Non-Contradiction	$\not\Vdash \sim t$ when $t \in O$	**Min-$\forall$**	$\Vdash [\![\forall v \phi]\!]_g$ iff $\Vdash [\![\phi]\!]_{g[v \mapsto p]}$ for all $p \in \mathcal{R}(S, \sqsubseteq)$
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Exploring the Theory of RU-Models

Let's consider which of the axioms of B+ are true in all RU-models. We get all but five of the Boolean identities:

Com	$p \wedge q = q \wedge p$	$p \vee q = q \vee p$
Assoc	$p \wedge (q \wedge r) = (p \wedge q) \wedge r$	$p \vee (q \vee r) = (p \vee q) \vee r$
Id	$p \wedge p = p$	$p \vee p = p$
Dist	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
Abs	$\cancel{p \wedge (p \vee q)} = \cancel{p}$	$\cancel{p \vee (p \wedge q)} = \cancel{p}$
Comp	$\cancel{p \wedge \neg p} = \cancel{q \wedge \neg q}$	$\cancel{p \vee \neg p} = \cancel{q \vee \neg q}$
DeM	$\neg(p \vee q) = \neg p \wedge \neg q$	$\neg(p \wedge q) = \neg p \vee \neg q$
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Comp	$\cancel{p \wedge \neg p} = \cancel{q \wedge \neg q}$	$\cancel{p \vee \neg p} = \cancel{q \vee \neg q}$
DeM	$\neg(p \vee q) = \neg p \wedge \neg q$	$\neg(p \wedge q) = \neg p \vee \neg q$
Invol	$\cancel{\neg\neg p} = \cancel{p}$	

Call the result of adding these equations to classical logic **A** (after R.B. Angell).

Exploring the theory of RU-models

We also get to keep all the "+" principles, but with two slight weakenings:

Weak $\vee$-**Comp**	$\exists p \gamma[p] \to \exists p\, \mathbf{Sup}_{\leq_\vee}(p, \gamma)$
Weak $\wedge$-**Comp**	$\exists p \gamma[p] \to \exists p\, \mathbf{Sup}_{\leq_\wedge}(p, \gamma)$
Atomicity	$\forall q(p \leq_\vee q) \vee \exists q(\mathbf{Atom}_{\leq_\vee} q \wedge q \leq_\vee p)$
Atomisticity	$\mathbf{Sup}_{\leq_\vee}(p, \mathbf{Atom}_{\leq_\vee} \hat{q} \wedge \hat{q} \leq_\vee p)$
Actuality	$\exists p(p \wedge \forall q(q \to q \leq_\wedge p))$
Anti-Actuality	$\exists p(\neg p \wedge \forall q(\neg q \to q \leq_\vee p))$
True Atom	$\exists p(\mathbf{Atom}_{\leq_\vee} p \wedge p)$

Call this theory **A+**.

This list is somewhat redundant (e.g., Atomicity follows in A from Atomisticity, and True Atom follows from Anti-Actuality and Atomisticity), though much less redundant than in B.

The theory Internal RU

$\mathcal{L}_{\mathbf{TM}}$ extends $\mathcal{L}$ with *state variables* $s, t, u, \ldots$.

When ϕ is a formula and $s, t, \ldots$ are state variables, the following are also formulae:

$s \rhd \phi$ ("s verifies ϕ")	$\sim s$
$s \sqsubseteq t$ ("s is contained in/a part of t")	$\forall s \phi$
$\mathbf{Obt}\, s$ ("s obtains")	$\exists s \phi$

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Abbreviations: $s \rhd_{\sqsubseteq} \phi \coloneqq \exists t(t \rhd \phi \wedge t \sqsubseteq s)$

$$\mathbf{Reg}(\gamma) \coloneqq \forall s(\gamma[s] \leftrightarrow \exists t \exists u(\gamma[t] \wedge \mathbf{Sup}_{\sqsubseteq}(u, \gamma) \wedge t \sqsubseteq s \wedge s \sqsubseteq u)$$

Internal RU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$

Internal RU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \to \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \to t \sqsubseteq s))$

Internal RU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Ob}\, t\, s \wedge t \sqsubseteq s \to \mathbf{Ob}\, t\, t$	**I-Ob Closure**	$\exists s(\mathbf{Ob}\, t\, s \wedge \forall t(\mathbf{Ob}\, t\, t \to t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Ob}\, t\, t \to \neg \mathbf{Ob}\, t\, s$	**I-Biv**	$\neg \mathbf{Ob}\, t\, s \to \exists t(\mathbf{Ob}\, t\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \to \mathbf{Sup}_{\leq_{\vee}}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t))$		

Internal RU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \rightarrow s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \rightarrow s = t$	**I-Comp**	$\exists s \gamma[s] \rightarrow \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \rightarrow \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \rightarrow t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim}t \wedge \mathbf{Obt}\, t \rightarrow \neg \mathbf{Obt}\, s$	**I-Biv**	$\neg \mathbf{Obt}\, s \rightarrow \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim}s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \rightarrow \mathbf{Sup}_{\leq_\vee}({\sim}s, \exists t(t \rhd p \wedge \hat{x} = {\sim}t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \rightarrow \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \rightarrow p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		

Internal RU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \to \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \to t \sqsubseteq s))$
I-NonCon	$s \triangleright {\sim} t \wedge \mathbf{Obt}\, t \to \neg\, \mathbf{Obt}\, s$	**I-Biv**	$\neg\, \mathbf{Obt}\, s \to \exists t(\mathbf{Obt}\, t \wedge t \triangleright {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \triangleright p) \to \mathbf{Sup}_{\leq_\vee}({\sim} s, \exists t(t \triangleright p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \triangleright p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \to \exists p \forall s(s \triangleright p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \triangleright p$	**I-Uniqueness**	$\forall s(s \triangleright p \leftrightarrow s \triangleright q) \to p = q$
I-Truth	$p \leftrightarrow \exists s(s \triangleright p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \triangleright \neg p \leftrightarrow \exists q(s \triangleright q \wedge \mathbf{Sup}_{\leq_\wedge}(q, \exists t(t \triangleright p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \triangleright (p \wedge q) \leftrightarrow (s \triangleright_{\sqsubseteq} p \wedge s \triangleright_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \triangleright p \vee \hat{x} \triangleright q) \wedge s \sqsubseteq t)$		
I-Disj	$s \triangleright (p \vee q) \leftrightarrow (s \triangleright_{\sqsubseteq} p \vee s \triangleright_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, x \triangleright p \vee x \triangleright q) \wedge s \sqsubseteq t)$		

The reduction

Strategy: use singleton subsets of S as surrogates for states. As in the PW case, we can pick out singletons in $\mathcal{L}$: $\mathbf{Atom}_{\leq_\vee}(\phi)$ is true on g iff $[\![\phi]\!]_g = \{s\}$ for some s.

We define a mapping $|\cdot|$ from $\mathcal{L}_{\mathbf{TM}}$ to $\mathcal{L}$:

$$|s \sqsubseteq t| := s \leq_\wedge t \qquad |{\sim}s| := \neg s$$

$$|s \rhd \phi| := s \leq_\vee |\phi| \qquad |\mathbf{Obt}\, s| := s$$

$$|\forall s \phi| := \forall s(\mathbf{Atom}_{\leq_\vee} s \to |\phi|) \qquad |\exists s \phi| := \exists s(\mathbf{Atom}_{\leq_\vee} s \wedge |\phi|)$$

$$|\phi \wedge \psi| := |\phi| \wedge |\psi| \qquad |\phi \vee \psi| := |\phi| \vee |\psi|$$

$$|\neg \phi| := \neg|\phi| \qquad |\phi = \psi| := |\phi| = |\psi|$$

$$|\forall p \phi| := \forall p|\phi| \qquad |\exists p \phi| := \exists p|\phi|$$

Fact

$|\phi|$ is true in every RU-model whenever ϕ is a theorem of Internal RU.

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \to \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \to t \sqsubseteq s))$
I-NonCon	$s \triangleright {\sim} t \wedge \mathbf{Obt}\, t \to \neg\, \mathbf{Obt}\, s$	**I-Biv**	$\neg\, \mathbf{Obt}\, s \to \exists t(\mathbf{Obt}\, t \wedge t \triangleright {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \triangleright p) \to \mathbf{Sup}_{\leq_{\vee}}({\sim} s, \exists t(t \triangleright p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \triangleright p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \to \exists p \forall s(s \triangleright p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \triangleright p$	**I-Uniqueness**	$\forall s(s \triangleright p \leftrightarrow s \triangleright q) \to p = q$
I-Truth	$p \leftrightarrow \exists s(s \triangleright p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \triangleright \neg p \leftrightarrow \exists q(s \triangleright q \wedge \mathbf{Sup}_{\leq_{\wedge}}(q, \exists t(t \triangleright p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \triangleright (p \wedge q) \leftrightarrow (s \triangleright_{\sqsubseteq} p \wedge s \triangleright_{\sqsubseteq} q) \wedge \exists t\, (\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \triangleright p \vee \hat{x} \triangleright q) \wedge s \sqsubseteq t)$		
I-Disj	$s \triangleright (p \vee q) \leftrightarrow (s \triangleright_{\sqsubseteq} p \vee s \triangleright_{\sqsubseteq} q) \wedge \exists t\, (\mathbf{Sup}_{\sqsubseteq}(t, x \triangleright p \vee x \triangleright q)) \wedge s \sqsubseteq t)$		

From classical logic on its own, we get the $|\cdot|$-translations of:

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \rightarrow s \sqsubseteq u$	I-Reflex	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \rightarrow s = t$	I-Comp	$\exists s \gamma[s] \rightarrow \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \rightarrow \mathbf{Obt}\, t$	I-Ob Closure	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \rightarrow t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Obt}\, t \rightarrow \neg\, \mathbf{Obt}\, s$	I-Biv	$\neg\, \mathbf{Obt}\, s \rightarrow \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \rightarrow \mathbf{Sup}_{\leq_\vee}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	I-Existence	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \rightarrow \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	I-Uniqueness	$\forall s(s \rhd p \leftrightarrow s \rhd q) \rightarrow p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \rhd \neg p \leftrightarrow \exists q(s \rhd q \wedge \mathbf{Sup}_{\leq_\wedge}(q, \exists t(t \rhd p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \rhd (p \wedge q) \leftrightarrow (s \rhd_{\sqsubseteq} p \wedge s \rhd_{\sqsubseteq} q) \wedge \exists t\, (\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \rhd p \vee \hat{x} \rhd q) \wedge s \sqsubseteq t)$		
I-Disj	$s \rhd (p \vee q) \leftrightarrow (s \rhd_{\sqsubseteq} p \vee s \rhd_{\sqsubseteq} q) \wedge \exists t\, (\mathbf{Sup}_{\sqsubseteq}(t, x \rhd p \vee x \rhd q)) \wedge s \sqsubseteq t)$		

From classical logic and A, we get the $|\cdot|$-translations of:

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \to \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \to t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim}t \wedge \mathbf{Obt}\, t \to \neg\, \mathbf{Obt}\, s$	**I-Biv**	$\neg\, \mathbf{Obt}\, s \to \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim}s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \to \mathbf{Sup}_{\leq_\vee}({\sim}s, \exists t(t \rhd p \wedge \hat{x} = {\sim}t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \to \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \to p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \rhd \neg p \leftrightarrow \exists q(s \rhd q \wedge \mathbf{Sup}_{\leq_\wedge}(q, \exists t(t \rhd p \wedge \hat{x} = {\sim}t)))$		
I-Conj	$s \rhd (p \wedge q) \leftrightarrow (s \rhd_{\sqsubseteq} p \wedge s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \rhd p \vee \hat{x} \rhd q) \wedge s \sqsubseteq t)$		
I-Disj	$s \rhd (p \vee q) \leftrightarrow (s \rhd_{\sqsubseteq} p \vee s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, x \rhd p \vee x \rhd q)) \wedge s \sqsubseteq t)$		

From classical logic and A+, we get the $|\cdot|$-translations of:

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \rightarrow s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \rightarrow s = t$	**I-Comp**	$\exists s \gamma[s] \rightarrow \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \rightarrow \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \rightarrow t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Obt}\, t \rightarrow \neg \mathbf{Obt}\, s$	**I-Biv**	$\neg \mathbf{Obt}\, s \rightarrow \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \rightarrow \mathbf{Sup}_{\leq_{\vee}}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \rightarrow \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \rightarrow p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \rhd \neg p \leftrightarrow \exists q(s \rhd q \wedge \mathbf{Sup}_{\leq_{\wedge}}(q, \exists t(t \rhd p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \rhd (p \wedge q) \leftrightarrow (s \rhd_{\sqsubseteq} p \wedge s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \rhd p \vee \hat{x} \rhd q) \wedge s \sqsubseteq t)$		
I-Disj	$s \rhd (p \vee q) \leftrightarrow (s \rhd_{\sqsubseteq} p \vee s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, x \rhd p \vee x \rhd q)) \wedge s \sqsubseteq t)$		

Deriving Internal U-TM

More of the principles of Internal U-TM follow when we throw in the following infinitary analogues of principles we have already seen:

Inf $\vee\wedge$ **Dist**	$\mathbf{Sup}_{\leq_\wedge}(p, \gamma) \rightarrow \mathbf{Sup}_{\leq_\wedge}(p \vee q, \exists u\,(\gamma[u] \wedge \hat{x} = u \vee q)$
Inf $\wedge\vee$ **Dist**	$\mathbf{Sup}_{\leq_\vee}(p, \gamma) \rightarrow \mathbf{Sup}_{\leq_\vee}(p \wedge q, \exists u\,(\gamma[u] \wedge \hat{x} = u \wedge q)$
Inf $\wedge\vee$ **DeMorgan**	$\mathbf{Sup}_{\leq_\wedge}(p, \gamma) \rightarrow \mathbf{Sup}_{\leq_\vee}(\neg p, \exists u(\gamma[u] \wedge \hat{x} = \neg u)$
Inf $\vee\wedge$ **DeMorgan**	$\mathbf{Sup}_{\leq_\vee}(p, \gamma) \rightarrow \mathbf{Sup}_{\leq_\wedge}(\neg p, \exists u(\gamma[u] \wedge \hat{x} = \neg u)$

These are theorems of B^+, but are not obviously derivable in A^+.

Let A^{++} be the the theory that adds these to A^+.

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Ob}\, t\, s \wedge t \sqsubseteq s \to \mathbf{Ob}\, t\, t$	**I-Ob Closure**	$\exists s(\mathbf{Ob}\, t\, s \wedge \forall t(\mathbf{Ob}\, t\, t \to t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Ob}\, t\, t \to \neg\, \mathbf{Ob}\, t\, s$	**I-Biv**	$\neg\, \mathbf{Ob}\, t\, s \to \exists t(\mathbf{Ob}\, t\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \to \mathbf{Sup}_{\leq_\vee}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \to \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \to p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Ob}\, t\, s)$		
I-Neg	$s \rhd \neg p \leftrightarrow \exists q(s \rhd q \wedge \mathbf{Sup}_{\leq_\wedge}(q, \exists t(t \rhd p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \rhd (p \wedge q) \leftrightarrow (s \rhd_{\sqsubseteq} p \wedge s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \rhd p \vee \hat{x} \rhd q) \wedge s \sqsubseteq t)$		
I-Disj	$s \rhd (p \vee q) \leftrightarrow (s \rhd_{\sqsubseteq} p \vee s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, x \rhd p \vee x \rhd q)) \wedge s \sqsubseteq t)$		

In A^{++} we can derive the $|\cdot|$-translations of four more axioms:

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \rightarrow s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \rightarrow s = t$	**I-Comp**	$\exists s \gamma[s] \rightarrow \exists s \, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \rightarrow \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \rightarrow t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Obt}\, t \rightarrow \neg \mathbf{Obt}\, s$	**I-Biv**	$\neg \mathbf{Obt}\, s \rightarrow \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \rightarrow \mathbf{Sup}_{\leq_{\vee}}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t))$		
I-Regularity	$\mathbf{Reg}(\hat{s} \rhd p)$	**I-Existence**	$\mathbf{Reg}(\gamma) \wedge \exists s \gamma[s] \rightarrow \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \rightarrow p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		
I-Neg	$s \rhd \neg p \leftrightarrow \exists q(s \rhd q \wedge \mathbf{Sup}_{\leq_{\wedge}}(q, \exists t(t \rhd p \wedge \hat{x} = {\sim} t)))$		
I-Conj	$s \rhd (p \wedge q) \leftrightarrow (s \rhd_{\sqsubseteq} p \wedge s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, \hat{x} \rhd p \vee \hat{x} \rhd q) \wedge s \sqsubseteq t)$		
I-Disj	$s \rhd (p \vee q) \leftrightarrow (s \rhd_{\sqsubseteq} p \vee s \rhd_{\sqsubseteq} q) \wedge \exists t\,(\mathbf{Sup}_{\sqsubseteq}(t, x \rhd p \vee x \rhd q)) \wedge s \sqsubseteq t)$		

To derive the remaining principles, we require something more, such as:

Bottomlessness $\neg\exists p \forall q\, p \leq_\vee q$

Confluence $\exists r(p \leq_\wedge r \wedge q \leq_\vee r) \to \exists r(r \leq_\vee p \wedge r \leq_\wedge q)$

Disjunction Decomposition

$$p \leq_\vee (q \vee r) \to \exists p' \leq_\vee p\big(\big(p' \leq_\vee q \vee (q \wedge r)\big) \vee \big(p' \leq_\vee r \vee (q \wedge r)\big)\big)$$

All three of these are valid in RU-models, and indeed follow from Internal RU.

Let $\mathbf{A}^{+++}$ be the theory that adds these to $\mathbf{A}^{++}$.

To derive the remaining principles, we require something more, such as:

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Disjunction Decomposition

$$p \leq_\vee (q \vee r) \to \exists p' \leq_\vee p \Big(\big(p' \leq_\vee q \vee (q \wedge r) \big) \vee \big(p' \leq_\vee r \vee (q \wedge r) \big) \Big)$$

All three of these are valid in RU-models, and indeed follow from Internal RU.

Let $\mathbf{A}^{+++}$ be the theory that adds these to $\mathbf{A}^{++}$.

Theorem

$\mathbf{A}^{+++}$ is equivalent to $|\text{Internal RU}|$.

The Theory $\mathbf{A}^{+++}$

To review: $\mathbf{A}^{+++}$ comprises:

- Classical logic
- Angell Identities other than Involution (Idempotence, Associativity, Commutativity, Distribution, DeMorgan)
- Infinitary analogues of classical logic (Weak $\wedge$-Comp, Weak $\vee$-Comp, Sup-Truth)
- Infinitary Analogues of Angell Identities
- Atomisticity
- Bottomlessness (the only axiom which is not a theorem of B+).
- Confluence
- Disjunction Decomposition

- Classical Logic
- Angell Identities other than Involution (Idempotence, Associativity, Commutativity, Distribution, DeMorgan)
- Infinitary analogues of classical logic (Weak $\wedge$-Comp, Weak $\vee$-Comp, Actuality)
- Infinitary Analogues of Angell Identities
- Atomisticity
- Bottomlessness
- Confluence: $\exists r(p \leq_\wedge r \wedge q \leq_\vee r) \rightarrow \exists r(r \leq_\vee p \wedge r \leq_\wedge q)$
- Disjunction Decomposition:
  $p \leq_\vee (q \vee r) \rightarrow \exists r \leq_\vee p\,(r \leq_\vee q \vee (q \wedge r) \vee r \leq_\vee r \vee (q \wedge r))$

Regular, bilateral models

The Harder Bilateral Case

Let's turn to (regular, nonvacuous) bilateral models, where propositions are represented by pairs of regular nonempty sets of states.

To do something analogous in this setting to what we did in the unilateral setting, we'll need some notion of "the full domain of propositions" that applies to bilateral models.

The Harder Bilateral Case

Let's turn to (regular, nonvacuous) bilateral models, where propositions are represented by pairs of regular nonempty sets of states.

To do something analogous in this setting to what we did in the unilateral setting, we'll need some notion of "the full domain of propositions" that applies to bilateral models.

The first thing we tried exploited the notion of "consistency" for states: $\langle p, q \rangle$ is in the domain iff (i) no member of p is consistent with any member of q and (ii) for every consistent s there is some consistent s^+ that contains either a member of p or a member of q.

- We figured out how to do the "internalization" in this case. But Kit doubts that these models are metaphysically realistic.

Constraining the domain of bilateral propositions

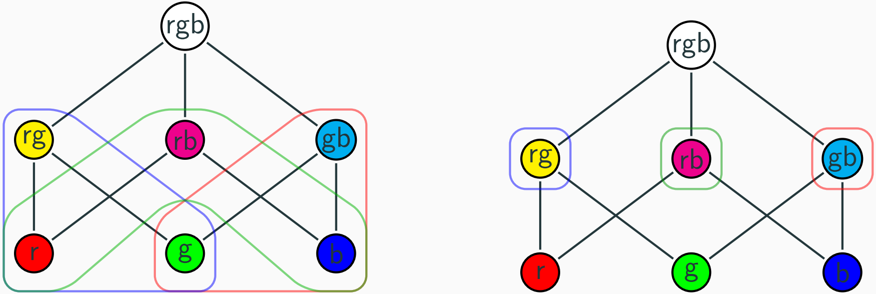
To have any hope of finding surrogates for states, we'll have to require the domain to be well-behaved in certain ways. Fine suggests keeping $\sim$ as part of the definition of model, and requiring:

Boolean Generation D is the closure under arbitrary conjunction, disjunction, and negation of the set $\{\langle\{s\}, {\sim}s\rangle \mid s \in S\}$.

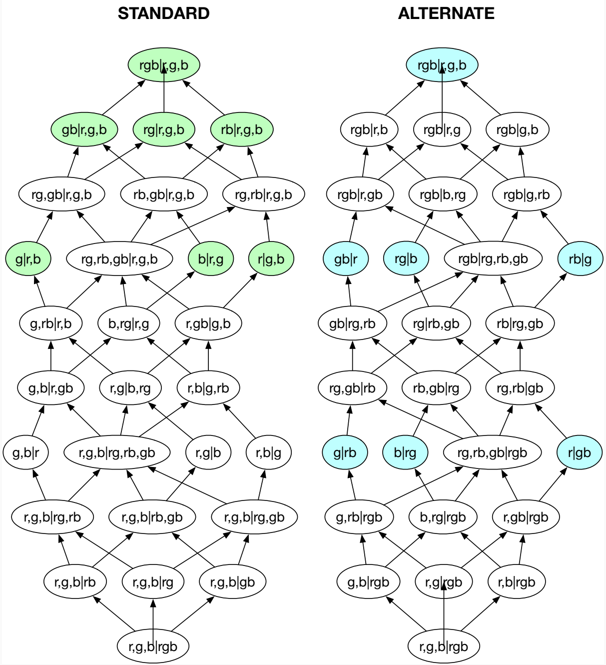
- Now we have natural surrogates for the states: the propositions $\langle\{s\}, {\sim}s\rangle$.
- But how can we pick them out in the object language? Without further constraints, there is no guarantee that they are disjunctive atoms. Also there may be other disjunctive atoms not of this form.

Impossibility result

Recall the following two ~-structures on the RGB state space:



The two bilateral models derived from these exclusion relations by Boolean Generation have isomorphic propositional domains, but the isomorphism maps propositions with exactly one verifier to propositions with multiple verifiers.


STANDARD
ALTERNATE
rgb|r,g,b
gb|r,g,b
rg|r,g,b
rb|r,g,b
rg,gb|r,g,b
rb,gb|r,g,b
rg,rb|r,g,b
g|r,b
rg,rb,gb|r,g,b
b|r,g
r|g,b
g,rb|r,b
b,rg|r,g
r,gb|g,b
g,b|r,gb
r,g|b,rg
r,b|g,rb
g,b|r
r,g,b|rg,rb,gb
r,g|b
r,b|g
r,g,b|rg,rb
r,g,b|rb,gb
r,g,b|rg,gb
r,g,b|rb
r,g,b|rg
r,g,b|gb
r,g,b|rgb
rgb|r,g,b
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rg,rb,gb|rgb
r|gb
g,rb|rgb
b,rg|rgb
r,gb|rgb
g,b|rgb
r,g|rgb
r,b|rgb
r,g,b|rgb

Note: these models actually have 76 propositions each. The previous slide depicts not propositions but equivalence classes of propositions under “L-equivalence”, i.e. having the same upward closure on both sides.

What we can do

Suppose we impose the following further constraints:

Separativity $\langle S, \sqsubseteq\rangle$ is separative (obeys Weak Supplementation): if $s \not\sqsubseteq t$, then there exists $s' \sqsubseteq s$ such that there does not exist u with $u \sqsubseteq t$ and $u \sqsubseteq s'$.

Converse Exclusion $s \in \dot{\bigwedge}\{{\sim}t : t \in {\sim}s\}$

Downward Exclusion $({\sim}s)^{\downarrow} = {\sim}s$

M-**Definiteness** If $\langle\{s\}, p\rangle \in D$ and $\langle\{s\}, q\rangle \in D$, then $\sup p = \sup q$.

- Note that these constraints guarantee that $p \subseteq {\sim}s$ whenever $\langle\{s\}, p\rangle \in D$, so we can recover $\sim$ from D.

What we can do

Define:

$$\mathbf{State}\,\phi \coloneqq \mathbf{Atom}_{\leq_\vee}\,\phi \wedge \forall q(q \leq_L \phi \rightarrow \phi \leq_\wedge q)$$

Theorem

In bilateral models satisfying Boolean Generation, Separativity, Converse Exclusion, Downward Exclusion, and M-definiteness, $\mathbf{State}\,\phi$ is true exactly when $[\![\phi]\!] = \langle \{s\}, {\sim}s \rangle$ for some s.

Thus **State** can play the same role in the internalization that we did with **Atom** in the unilateral case. With more work, we can find appropriate translations of "verifies", "falsifies", etc.

Irregular unilateral truthmaker models

Irregular, unilateral truthmaker models

$\mathcal{P}^+(S) \coloneqq \{p \subseteq S : p \text{ is non-empty}\}$

A **IU-model** is a tuple $\mathfrak{M} = \langle S, \sqsubseteq, O, \sim, [\![\cdot]\!] \rangle$, where $\langle S, \sqsubseteq \rangle$ is a complete semilattice, $O \subseteq S$, , and $[\![\cdot]\!]^{\cdot} : \mathcal{P}^+(S)^{\mathbf{Var}} \to \mathcal{P}^+(S)^{\mathcal{L}}$, such that:

		Variables	$[\![p]\!]_g = g(p)$
Obtaining Parts	$O = O^{\downarrow}$	**Conjunction**	$[\![\phi \wedge \psi]\!]_g = [\![\phi]\!]_g \,\dot{\sqcup}\, [\![\psi]\!]_g$
Obtaining Closure	$\sup O \in O$	**Disjunction**	$[\![\phi \vee \psi]\!]_g = [\![\phi]\!]_g \cup [\![\psi]\!]_g$
		Negation	$[\![\neg\phi]\!] = \dot{\bigsqcup}\{\sim s : s \in [\![\phi]\!]_g\}$

Non-Contradiction	$\nVdash \sim t$ when $t \in O$	**Min-$\forall$**	$\Vdash [\![\forall v \phi]\!]_g$ iff $\Vdash [\![\phi]\!]_{g[v \mapsto p]}$ for all $p \in \mathcal{P}^+(S)$
Bivalence	$\Vdash \sim t$ when $t \notin O$	**Min-$\exists$**	$\Vdash [\![\forall v \phi]\!]_g$ iff $\Vdash [\![\phi]\!]_{g[v \mapsto p]}$ for some $p \in \mathcal{P}^+(S)$
Sup Exclusion	$\sim \sup p = \left(\bigcup\{\sim s : s \in p\}\right)^*$	**Min-$=$**	$\Vdash [\![\phi = \psi]\!]_g$ iff $[\![\phi]\!]_g = [\![\psi]\!]_g$.
Closed Exclusion	$\sup p \in \sim s$ when $p \subseteq \sim s$		

$\Vdash p$ (p is **true in** $\mathfrak{M}$) iff $O \cap p \neq \varnothing$. ϕ is **true in** $\mathfrak{M}$ iff $\Vdash [\![\phi]\!]_g$ for all g.

Exploring the Theory of IU-Models

We lose two and a half more Boolean identities:

Com	$p \wedge q = q \wedge p$	$p \vee q = q \vee p$
Assoc	$p \wedge (q \wedge r) = (p \wedge q) \wedge r$	$p \vee (q \vee r) = (p \vee q) \vee r$
Id	$\cancel{p \wedge p} = \cancel{p}$	$p \vee p = p$
Dist	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	$\cancel{p \vee (q \wedge r)} = \cancel{(p \vee q) \wedge (p \vee r)}$
Abs	$\cancel{p \wedge (p \vee q)} = \cancel{p}$	$\cancel{p \vee (p \wedge q)} = \cancel{p}$
Comp	$\cancel{p \wedge \neg p} = \cancel{q \wedge \neg q}$	$\cancel{p \vee \neg p} = \cancel{q \vee \neg q}$
DeM	$\neg(p \vee q) = \neg p \wedge \neg q$	$\neg p \vee \neg q \leq_{\vee} \neg(p \wedge q)$
Invol	$\cancel{\neg\neg p} = \cancel{p}$	

Exploring the Theory of IU-Models

We lose two and a half more Boolean identities:

Com	$p \land q = q \land p$	$p \lor q = q \lor p$
Assoc	$p \land (q \land r) = (p \land q) \land r$	$p \lor (q \lor r) = (p \lor q) \lor r$
Id	$\cancel{p \land p} = \cancel{p}$	$p \lor p = p$
Dist	$p \land (q \lor r) = (p \land q) \lor (p \land r)$	$\cancel{p \lor (q \land r)} = \cancel{(p \lor q) \land (p \lor r)}$
Abs	$\cancel{p \land (p \lor q)} = \cancel{p}$	$\cancel{p \lor (p \land q)} = \cancel{p}$
Comp	$\cancel{p \land \neg p} = \cancel{q \land \neg q}$	$\cancel{p \lor \neg p} = \cancel{q \lor \neg q}$
DeM	$\neg(p \lor q) = \neg p \land \neg q$	$\neg p \lor \neg q \leq_{\lor} \neg(p \land q)$
Invol	$\cancel{\neg\neg p} = \cancel{p}$	

Call the result of adding these equations to classical logic $\mathbf{A}'$.

Exploring the theory of IU-models

We get the "+" principles, but with one further modification:

Weak $\vee$-**Comp**	$\exists p \gamma[p] \to \exists p\, \mathbf{Sup}_{\leq_\vee}(p, \gamma)$
Regular $\wedge$-**Comp**	$\exists p \gamma[p] \to \exists p\, \mathbf{Sup}_{\leq_\wedge}(p, \gamma, \mathbf{Reg}(\hat{q}))$
Atomicity	$\forall q(p \leq_\vee q) \vee \exists q(\mathbf{Atom}_{\leq_\vee} q \wedge q \leq_\vee p)$
Atomisticity	$\mathbf{Sup}_{\leq_\vee}(p, \mathbf{Atom}_{\leq_\vee} \hat{q} \wedge \hat{q} \leq_\vee p)$
Actuality	$\exists p(p \wedge \forall q(q \to q \leq_\wedge p))$
Anti-Actuality	$\exists p(\neg p \wedge \forall q(\neg q \to q \leq_\vee p))$
True Atom	$\exists p(\mathbf{Atom}_{\leq_\vee} p \wedge p)$

where

$$\mathbf{Reg}(\phi) \coloneqq \mathbf{Convex}(\phi) \wedge \mathbf{ContainsSup}(\phi)$$

$$\mathbf{Convex}(\phi) \coloneqq \forall q_1 q_2 q_3 (q_1 \leq_\vee \phi \wedge q_3 \leq_\vee \phi \wedge q_1 \leq_\wedge q_2 \wedge q_2 \leq_\wedge q_3 \to q_2 \leq_\vee \phi)$$

$$\mathbf{ContainsSup}(\phi) \coloneqq \exists p(\mathbf{Atom}_{\leq_\vee} p \wedge p \leq_\vee \phi \wedge \phi \leq_\wedge p)$$

Exploring the theory of IU-models

We also get modified versions of some further principles characteristic of $\mathbf{A}^{+++}$:

Inf $\wedge\vee$ Dist $\qquad \mathbf{Sup}_{\leq_\vee}(p,\gamma) \to \mathbf{Sup}_{\leq_\vee}(p \wedge q, \exists u\,(\gamma[u] \wedge \hat{x} = u \wedge q))$

Bottomlessness $\qquad \neg\exists p\,\forall q\, p \leq_\vee q$

Confluence$_\exists$ $\qquad \exists r(p \leq_{\wedge\exists} r \wedge q \leq_\vee r) \to \exists r(r \leq_\vee p \wedge r \leq_{\wedge\exists} q)$

Disjunction Decomposition

$$p \leq_\vee (q \vee r) \to \exists p' \leq_\vee p\Big(\big(p' \leq_\vee q \vee (q \wedge r)\big) \vee \big(p' \leq_\vee r \vee (q \wedge r)\big)\Big)$$

Inf $\vee\wedge$ Dist fails for the same reason as $\vee\wedge$ Dist.

Let A'^{+++} be the the theory that adds these to A'^{+}.

The theory Internal IU

I-Trans	$s \sqsubseteq t \wedge t \sqsubseteq u \rightarrow s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \wedge t \sqsubseteq s \rightarrow s = t$	**I-Comp**	$\exists s \gamma[s] \rightarrow \exists s\, \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \wedge t \sqsubseteq s \rightarrow \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \wedge \forall t(\mathbf{Obt}\, t \rightarrow t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \wedge \mathbf{Obt}\, t \rightarrow \neg\, \mathbf{Obt}\, s$	**I-Biv**	$\neg\, \mathbf{Obt}\, s \rightarrow \exists t(\mathbf{Obt}\, t \wedge t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \rightarrow \mathbf{Sup}_{\leq_\vee}({\sim} s, \exists t(t \rhd p \wedge \hat{x} = {\sim} t), \mathbf{Cl}(\hat{q}))$		
I-Closed Excl	$\forall t(\gamma[t] \rightarrow t \rhd {\sim} s) \wedge \mathbf{Sup}_{\sqsubseteq}(t', \gamma) \rightarrow t' \rhd {\sim} s$		
I-Existence	$\exists s \gamma[s] \rightarrow \exists p \forall s(s \rhd p \leftrightarrow \gamma[s])$		
I-Nonempty	$\exists s\, s \rhd p$	**I-Uniqueness**	$\forall s(s \rhd p \leftrightarrow s \rhd q) \rightarrow p = q$
I-Truth	$p \leftrightarrow \exists s(s \rhd p \wedge \mathbf{Obt}\, s)$		
I-Conj′	$s \rhd (p \wedge q) \leftrightarrow \exists t \exists t'(t \rhd p \wedge t' \rhd q \wedge s = t \sqcup t')$		
I-Disj′	$s \rhd (p \vee q) \leftrightarrow (s \rhd p \vee s \rhd q)$		
I-Neg′	$s \rhd \neg p \leftrightarrow \exists q(\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd q) \wedge \forall s'(s' \rhd q \rightarrow \exists t(t \rhd p \wedge s' \rhd {\sim} t)) \wedge \forall s'(s' \rhd p \rightarrow \exists t(t \rhd q \wedge s' \rhd {\sim} t)))$		

Reducing Internal IU

We apply the translation-scheme $|\cdot|$ as in the regular case.

However, not all of the principles of Internal IU are mapped by $|\cdot|$ to theorems of A'^{+++}—we need something further.

Reducing Internal IU

We apply the translation-scheme $|\cdot|$ as in the regular case.

However, not all of the principles of Internal IU are mapped by $|\cdot|$ to theorems of A'^{+++}—we need something further.

Fancy Confluence $\exists r(p \leq_\vee r \wedge q \leq_\wedge r) \to \exists r \exists r'(r \leq_\vee p \wedge r' \leq_\vee q \wedge r' \leq_\wedge r)$

Conjunction Decomposition $p \leq_\vee (q \wedge r) \to \exists q' \exists r'(q' \leq_\vee q \wedge r' \leq_\vee r \wedge q' \wedge r' \leq_\vee p)$

Strong Disjunction Decomposition $p \leq_\vee (q \vee r) \to \exists p' \leq_\vee p(p' \leq_\vee q \vee p' \leq_\vee r)$

I-Trans	$s \sqsubseteq t \land t \sqsubseteq u \to s \sqsubseteq u$	**I-Reflex**	$s \sqsubseteq s$
I-Antisym	$s \sqsubseteq t \land t \sqsubseteq s \to s = t$	**I-Comp**	$\exists s \gamma[s] \to \exists s \mathbf{Sup}_{\sqsubseteq}(s, \gamma)$
I-Ob Parts	$\mathbf{Obt}\, s \land t \sqsubseteq s \to \mathbf{Obt}\, t$	**I-Ob Closure**	$\exists s(\mathbf{Obt}\, s \land \forall t(\mathbf{Obt}\, t \to t \sqsubseteq s))$
I-NonCon	$s \rhd {\sim} t \land \mathbf{Obt}\, t \to \neg \mathbf{Obt}\, s$	**I-Biv**	$\neg \mathbf{Obt}\, s \to \exists t(\mathbf{Obt}\, t \land t \rhd {\sim} s)$
I-Sup Excl	$\mathbf{Sup}_{\sqsubseteq}(s, \hat{t} \rhd p) \to \mathbf{Sup}_{\leq_\vee}({\sim} s, \exists t(t \rhd p \land \hat{x} = {\sim} t), \mathbf{Cl}(\hat{q}))$		
I-Closed Excl	$\forall t(\gamma[t] \to t \rhd {\sim} s) \land \mathbf{Sup}_{\sqsubseteq}(t', \gamma) \to t' \rhd {\sim} s$		
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I-Truth	$p \leftrightarrow \exists s(s \rhd p \land \mathbf{Obt}\, s)$		
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